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GOODS AND SERVICES CONTENT SELECTION BASED ON ENVIRONMENT SCANS

Abstract

The present disclosure is directed to systems and methods for selecting objects and content items related to a given environment. In some embodiments, the systems and methods receive a mapping of the environment and extract layout, available space, and existing object data of the environment from the mapping. The systems and methods receive a storage of supplement content items, and select a supplemental content item from the storage of content items based on the received mapping. The systems and methods may further select the supplemental content item based on a user profile received from a data source. The systems and methods then generate for display the supplemental content item on a device to a user.

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Background/Summary

BACKGROUND

[0001] The present disclosure is related to systems and methods for scanning and analyzing a real-world environment to present recommendations, such as via unique content items, for enhancing use of the environment. The systems and methods may further select and recommend objects and/or services for better utilizing or making improvements to the environment.

SUMMARY

[0002] Updating or enhancing an environment (e.g., one or more rooms in a home) is generally a quite manual process that depends on a user to identify areas for modification or improvement, and to find the solutions, or products, him or herself. Systems and applications intended to assist in visualizing changes to an environment do exist, such as those in which a user may select a product and, using AR, project a realistic image of that product into the environment, e.g., overlaying an image of a television in a room. However, such AR applications generally depend on the user initially discovering the products to place (e.g., the user selects a television to place in the room), and identifying a particular location in the environment to place the product (e.g., the user wishes to place a television in an open space on an entertainment center).

[0003] There also exists applications for mapping real world environments, including objects within the environment, for the purpose of creating a representation of the environment. A robotic vacuum, for instance, may develop a mapping of a room to determine where to vacuum. Such applications, however, are not designed to assess the spatial information and other relevant information regarding the real-world environment to make recommendations that would help to improve or enhance the environment.

[0004] The present disclosure relates to systems and methods for analyzing various types of environmental information to recommend and provide visualizations of improvements to real-world environments. For example, the systems and methods may analyze the representations and data to extract spatial information of the given environment. The present disclosure provides techniques for analyzing the available representations or data of an environment, and based at least in part on that analysis, selecting goods or services for the environment. After making a selection, a computing device may present, to a user, a content item depicting or otherwise representing the good or service. The content item may be, for example, an image or video displayed to a user on a user computing device, such as a smartphone or augmented reality device. In some embodiments the content item illustrates the improvement by, for example, displaying an image of a selected product in a representation of the environment itself. In some embodiments, the same user device displaying the content item may also provide at least some of the data of the environment. For example a VR headset may generate scans or images of the environment and produce a representation of the environment including a selected product therein.

[0005] The present disclosure describes systems and techniques for integrating environment data into a content item selection processes. Multi-dimensional scans and other spatial information are collected for various spaces. For example, an XR gaming headset might capture a user's space while preparing motion boundaries. In another example, existing software may generate three-dimensional renderings of a space based on images of that space on a user's phone. In another example, home electronics, such as a robotic vacuum cleaner may log spatial information on which a rendering may be based. In some embodiments, the present disclosure may collect and incorporate data from adjacent or nearby environments into the content item selection process. In some embodiments, a system may receive spatial information and create multi-dimensional renderings itself based on that data. In some embodiments, a system may receive already created renderings.

[0006] In one approach, the systems and techniques receive an inventory of available or offered products. The described systems and techniques then select, based on the spatial information or renderings, a product from the inventory that best suits the space. For example, the system might choose the product that best fills an empty space, that best matches a room's décor, that best replaces an existing object, and/or that best addresses a product absence. The system may also select or generate content items based on spatial information about nearby environments. For example, the system may determine that a specific product is not recommended for an area of the environment in a field of view based on information that another (perhaps nearby) area or field of view contains the same or similar product. In some embodiments the environment comprises a space beyond a present field of view, such as alternative fields of view of the same space, nearby rooms or spaces, or related spaces. Once the system selects the product or service, it may display a depiction of the product or service to the user as a content item. In one example, the content item shows an image of the product installed in the environment. The content may also be, for example, a three-dimensional object that can be sized and placed in an VR or AR environment to represent a real world item in the environment. In this way, the disclosed systems and techniques generate highly personalized content items unique to each user.

[0007] The disclosed personalized content selection system generates recommendations tailored to a specific environment without requiring user input, awareness, or intent with respect to potential improvements. It further creates a unique representation of the environment including the recommended products or services, and may take into account neighboring environments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 shows an example process in accordance with some embodiments of the present disclosure;

[0009] FIG. 2 shows an example architecture of some embodiments of the present disclosure;

[0010] FIG. 3 shows an example process of some embodiments of the present disclosure;

[0011] FIG. 4 shows an example three-dimensional mapping of some embodiments of the present disclosure;

[0012] FIG. 5 shows an example three-dimensional mapping of some embodiments of the present disclosure;

[0013] FIG. 6 shows an example of the object selection process of some embodiments of the present disclosure; and

[0014] FIG. 7 shows an example process in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 shows an illustrative embodiment of a content selection system **100** that displays content on a primary device based on spatial information collected at a secondary device. At a first step, a secondary device **101** collects spatial information of a three-dimensional, physical or real-world environment **102**, such as a user's living room. The device **101** may be an XR headset, smartphone, camera, or any other device capable of collecting spatial information of physical or real-world environments. Such devices may collect images or other data that, after analysis or processing, may inform a three dimensional rendering or mapping of the environment **102**. The content collection system **100** may analyze images, rendering, mapping, raw data, or other information to catalog locations or objects within the environment **102**. In this example, the content selection system **100** uses the collected spatial information to generate a mapping **103** of the environment **102** or otherwise analyze the environment **102**, as discussed in more detail below. The mapping **103** may document physical aspects of the environment **102** such as, for example,

dimensions, empty space, a layout, or a color scheme. The mapping may be a 3D rendering of the environment **102** as shown in FIG. **1**. In some embodiments, it is a 2D image, e.g., a floorplan or a mapping of a wall or other area. In some embodiments, it is data stored on the content selection system **100** that content selection system **100** may or may not represent to the user. The secondary device **101** may collect the spatial information by, for example, scanning the environment **102** or capturing images of the environment **102**. The secondary device **101** may be any device capable of these actions for mapping, such as an extended reality (XR) wearable device (e.g., a virtual reality (VR) or an augmented reality (AR) headset), a smartphone, tablet, or a house automation robot capable of two-dimensional scanning (e.g., an automated vacuum, or a robotic lawnmower, or a robot capable of three-dimensional scanning such as Amazon's Astro). The environment **102** may be any real-world area. For example, a room for which a user might purchase electronics (e.g., a television or speakers), furniture (e.g., a sofa or coffee table), or other items (e.g., art, decor, or fitness equipment). The content selection system **100** is configured to create or receive the mapping **103** and, based at least in part on the mapping, to select goods and/or services **104** that are adapted, needed, or desired for the environment **102**. In some embodiments, the content selection system **100** receives permission, such as from secondary device **101**, to use acquired scans or mapping **103**. The permission may be received via a setting or a pop-up icon requesting access to data.

[0016] In some embodiments, the content selection system **100** may scan or otherwise collect and store data on more than one more environment, such as a room, field of view of a house on a device **101**, or other enclosure or area. In such embodiments the various regions may inform the content selection system **100** of each other, building on knowledge to map a complete dwelling. For example, data on adjacent regions might inform the content collection system **100** about lighting in a room. One instance of this situation might occur if a room has no windows, but because it borders a bright room with large windows and a skylight, that same room receives sunlight throughout the day.

[0017] In some embodiments the content selection system **100** selects the content item representing the goods and/or services **104** from a storage of goods and services or content items related to the goods and/or services **104**. Content items **105** may be, for example, content illustrating the goods and/or services **104**, such an image of goods, an image of the results of a service, an advertisement for a good, a video review of the goods or services, a two or three-dimensional representation of a goods or service, etc. FIG. **1** shows content item **105** as an altered image of the environment **105** that includes a realistic representation of the goods and/or services **104**, here a TV, in the environment **102**. The content selection system **100** may alter the size position, rotation or coloring of a content item **105** for optimal presentation. For example, in the case of displaying a content item **105** on an AR device, the content selection system **100** may rotate and enlarge an overlay content item **105** to most realistically integrate into an existing environment **102**. The selection may be based on, for example, determining an available, underutilized, or empty area in the environment **102**.

[0018] In some embodiments, the content selection system **100** may, using mapping **103**, catalog the existing objects in the environment **102** to identify needs for or deficiencies of goods and/or services **104**. In some embodiments, the needs or deficiencies of an environment **102** are based on data on surrounding or related environments **102**, such as an adjacent or nearby room. The cataloging may include details or characteristics of each existing object cataloged, such as type of object, requirements for the object such as electricity or network connectivity, potential services related to the object, or make, model, year, color, style, dimensions, and price range of each existing object cataloged. In some embodiments, a third party or other resource provides a catalog. In some embodiments, the content collection system **100** builds a catalog from a collection of goods and services **104**. The content system **100** may collect data regarding the goods and services such as make, model, year, color, price, availability, etc. It may then store this data for use in selecting a content item **105**.

[0019] The content selection system **100** may then next use these object details or characteristics, as well as or in place of, inferring available area in environment **102**, to select goods and/or services **104** for the environment **102**. The goods and/or services **104** may be a subset of goods offered for sale, for example. These goods may include objects, such as televisions, furniture, artwork, or other items that are likely to be placed in an environment **102**. Services may include cleaning services, such as window washing or housekeeping, repair services, such as plumbing or woodwork, or services specifically related to an environment **102**, such as organizing for a large closet. In some embodiments, the content selection system **100** infers one or more available regions in environment **102** for the goods or services **104**. For example, content selection system **100** may identify an empty wall for a bookcase or an outdated rug to be replaced. In some embodiments, the content selection system **100** detects objects or regions in environment **102** that might be replaced based on inefficiency or other criteria. For example, the content collection system **100** might detect a bookcase and determine that the bookcase would be more efficient as a murphy bed. In another example, the content collection system **100**, might detect a couch that is out of style and that a replacement would better fit the aesthetic of the environment **102**.

[0020] The content selection system **100** may next create a content item **105**, such as an image shown in FIG. **1** displaying the goods and/or services **104** in a rendering of the environment **102**. The rendering may be based on the mapping **103** discussed above and any other available data on the environment **102**, in combination with data regarding three-dimensional properties of the goods and/or services **104**. The rendering may further include details such as realistic lighting of the environment **104** and goods and/or service **104** in the content item **105**. The rendering may be for example a virtual rendering in VR, a 2D unique generated image, or a view through AR. In some embodiments the content selection system **100** generates the content item **105** as an audio representation of the goods and services **104** in environment **102** either in place of or in addition to an image. In such an embodiment, the content selection system **100** may determine acoustic properties of an environment **102** from a mapping **103**. The content selection system **100** may also consider audio specifications of the goods and/or services **104** and incorporate those details into content item **105**. In some embodiments the content item **105** is a profile of a good or service where the profile is content that contains information dedicated to the good or service, such as an advertisement. For example, the content item **105** may be an advertisement that discusses the features of the goods and/or services **104**. Such a content item **105** may be generated according to a template in one example. In some embodiments, content selection system **100** may select an existing content item **105** from an object storage containing a collection of content items **105**, as opposed to generating a new or custom content item **105**. For example, the content selection system **100** may select an advertisement from a collection of advertisements in an advertisement store to display as content item **105**. The advertisement may correspond to a selected goods and/or service **104**.

[0021] The content selection system **100** transfers the content item **105** to a user device, the primary device, where the user device displays the content item **105**. In embodiments in which the content item **105** is an advertisement, the user device displays the content item **105** as an advertisement for a user to receive and potentially select. The user device may be any device which displays media such as a personal computer, smart phone, TV, tablet, XR device, or other devices. In some embodiments, the media is an advertisement. A user may select an advertisement when the user clicks on the advertisement or otherwise provides input of a selection. Selecting the advertisement may take the user to a product page or other portal offering more information or purchase opportunities. This content selection system **100** allows an advertisement platform, such as a browser, social network, or an e-commerce platform to infer goods and services needs of a user by using mapping and imaging information captured through their use of a secondary device **101**. In some embodiments, the content selection system **100** creates a unique image of environment **102** using mapping **103** and displays that image to the user. In some embodiments, the unique image

includes content item **105** which displays a good and/or service **104** in the environment **105**. In some embodiments, the display of content item **105** is an overlay generated over an image of environment **102**. Such embodiments might require that the content item **105** be resized and/or reoriented to properly represent what the good or service **104** will look like in the environment **102**. In some embodiments, content items **105** might further require that content system **100** further adjust shadows, highlights, and other external factors to create a realistic representation of a good or service **104**. In those embodiments, content selection system **100** further processes content item **105** to ensure the correct placement and appearance of the content item **105** in the environment **102**. In some embodiments, in which the content item **105** is related to a service, content selection system **100** may display a representation of environment **102** showing the service having been performed as a content item **105**. For example, if the content item represents a window cleaning service, the content selection system **100** might display an image of environment **102** with clean, sparkling windows as a content item **105**. In another example, if the content item represents a room cleaning service, the content selection system **100** might display a video showing a scan of environment **102** clean and neat, as a content item **105**. In another example, if the goods or service **104** generates a sound, such as is the case with a new speaker, the content selection system **100** might display an image of the environment **102** with the speaker and the sound of the speaker playing music as content item **105**. In some embodiments, the content selection system **100** may consider the acoustic qualities of a room determined by the mapping **103**, and reflect those qualities in the content item **105**.

[0022] FIG. 2 shows an example system architecture of content selection system **100**. Content selection system **100** includes server **220** which in some embodiments performs the primary functions of the disclosure such as those described below in FIGS. 3 and 7. Control circuitry **224** may be based on any suitable processing circuitry and includes control circuits and memory circuits, which may be disposed on a single integrated circuit or may be discrete components. As referred to herein, processing circuitry should be understood to mean circuitry based on at least one microprocessors, microcontrollers, digital signal processors, programmable logic devices, field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), system-on-chip (SoC), application-specific standard parts (ASSPs), indium phosphide (InP)-based monolithic integration and silicon photonics, non-classical devices, organic semiconductors, compound semiconductors, “More Moore” devices, “More than Moore” devices, cloud-computing devices, combinations of the same, or the like, and may include a multi-core processor (e.g., dual-core, quad-core, hexa-core, or any suitable number of cores). In some embodiments, processing circuitry may be distributed across multiple separate processors or processing units, for example, multiple of the same type of processing units (e.g., two Intel Core i9 processors) or multiple different processors (e.g., an Intel Core i7 processor and an Intel Core i9 processor). Some control circuits may be implemented in hardware, firmware, or software. Control circuitry **224** in turn includes communication circuitry **224a**, storage **224b** and processing circuitry **224c**. Control circuitry **224** may be utilized to execute or perform any or all the systems, methods, processes, and outputs of one or more of FIGS. 1-7, or any combination of steps thereof.

[0023] In some embodiments, control circuitry **224** executes instructions for an application stored in memory (e.g., storage **224b**). Specifically, control circuitry **224** may be instructed by the application to perform the functions discussed herein. In some embodiments, any action performed by control circuitry **224** may be based on instructions received from the application. For example, the application may be implemented as software or a set of and/or one or more executable instructions that may be stored in storage **224b** and executed by control circuitry **224**. The application may be a client/server application where only a server application resides on server **220**.

[0024] Content selection system **100** includes secondary device **210**, which may capture a scan and/or mapping **103** of a three-dimensional environment **102**. Secondary device **210** may be any

device with scanning capability, such as an XR headset, camera, or smartphone. In some embodiments, the scan may be produced from existing images that may be compiled to infer information regarding the environment **102**. Secondary device includes input/output circuitry **211** which may receive and transmit data as well as control circuitry **213** with similar features and capabilities of control circuitry **224**. The content selection system **100** also includes a primary device **212** for displaying images or other content items. The primary device **212** may be, for example, a smartphone, television, tablet, XR headset, or computer. In some embodiments, the primary and secondary device **212**, **210** are the same device where primary device **212** has scanning capabilities and secondary device **210** includes a user interface or display. Primary device includes input/output circuitry **214** which may receive and transmit data as well as control circuitry **215** with similar features and capabilities of control circuitry **224**. The primary and secondary devices **212**, **210** are connected via a network **201**. The network **201** may be any informational network, such as the Internet. The network **201** further connects the primary and secondary devices **212**, **210** to a server **220**. The server **220** receives the scan and/or mapping **103** from the secondary device **212** and processes this information in a processor **224**. In some embodiments, processor **224** analyzes a scan of environment **102** to create a mapping **103**. The server may also receive additional information from a database, the primary device **212**, or the secondary device **210**, such as information regarding a user's preferences, history, or any other relevant information. In some embodiments, information regarding a user's preferences, history, areas **102**, or other user data is stored in a user profile **228**. In some embodiments, the functions and processes of the server may take place on the devices **210** and/or **212** using, for example, processors and memories within those devices. The server **220**, via processor **224**, analyzes all information and selects, based on the information, an object or objects, such as goods and/or services **104** from the object storage **222**, which stores a collection of objects or content items **105** related to an object such as images or advertisements. In some embodiments, the system selects an object first and then, in response to that selection, selects or generates a content item related to the object. The server then may, in some embodiments, generate a content item including, for example, a selected image or advertisement, using processor **224**. The content selection system **100** then sends or causes to be displayed the generated content item **105** on the primary device **212**.

[0025] FIG. 3 shows a flow diagram of an example process **300** of content selection system **100**. The steps of process **300** may take place primarily on one device or may take place among several devices including, for example, a cloud server. The process **300** begins at step **302** when a secondary device, such as secondary device **210**, akin to secondary device **101**, acquires data, such as scans or images of a three-dimensional space, akin to environment **102**. The data may be acquired by the device itself using, for example, a camera, or may be received from another source. The secondary device **210** may acquire the data through, for example, an internal camera which captures images of the environment **102**. In some embodiments, secondary device acquires data and scans through distance detection, such as proximity detection, SLAM, LIDAR, or infrared sensors.

[0026] In some embodiments, the content selection system **100** may detect multiple scans for a particular environment **102**, a room for example, or may detect that the user spends an extended period of time in that particular location. For example, if the scan has been acquired by a robot, the object segmentation may extract the user or other individuals. Upon such detection, the content selection system **100** may rank content items for objects fitting that environment **102** higher than other objects fitting other locations also related to the user. By doing so, the content selection system **100** assumes that the user spends more time at that location than others and hence allocates more weight to content for objects related to that environment **102**. Additional weighting may take into account other parameters, such as the time of the scans as well. For example, a scan of an environment **102** taken at night, while normally might indicate that a room is not popular, might not assume that conclusion due to the fact that activity is low during certain hours.

[0027] A processor, such as processor **224**, then at step **304** converts the acquired data into a mapping **103** of the space. For example, processor **224** may create a mapping **103** that is an image or layout, such as a floor plan, of the environment **102** based on the acquired data. Processor may alternatively or additionally create three-dimensional models of the environment **102** based on the data. This step may take place, for example, at a server or a primary or secondary device **212**, **210**. [0028] At step **306** a processor (e.g., processor **224**) may segment the mapping **103** using semantic segmentation. The segmentation allows for identification of objects, color schemes, textures, layout of the environment, and regions of the environment, among other purposes. This step may also take place, for example, at a server or a primary or secondary device.

[0029] The processor **224** then at step **308** extracts and identifies objects, such as furniture, electronics, lighting, artwork, electrical outlets, windows, or other elements, within the environment **102** using the segmentation and object recognition processes. The content selection system **100** then may catalog the identified objects in environment **102** to inform the selection of content item **105**. For example, existing objects in environment **102** may inform the selection in that the content selection system **100** may avoid content items **105** that are redundant or unnecessary given the existing objects. In another example, the content selection system **100** may select content items **105** related to a need in the environment **102** based on the existing objects. For example, content selection system **100** in one embodiment might recognize a dining room but no chairs. The content selection system **100** may then select content items **105** promoting dining room chairs. The processor may also analyze the mapping **103** to determine additional information, such as color scheme, style, or budget. This additional information may further inform the selection process as goods and services **104** that match a color scheme, style, or budget are most desirable. In some embodiments, the content selection system **100** extracts the dominant color for each of the objects identified in the room, and these colors are converted into text to perform a metadata mapping with the goods and services. For example, the content selection system **100** may choose a content item **105** promoting a white couch for a room that has mostly white furniture.

[0030] To determine style information, the content selection system **100** may create or receive two-dimensional representations of an environment **102** based on data from one or a series of cameras. The data may contain color or texture information in some embodiments. An image-to-text model fine-tuned on style may then assess the representations to extract the style of the environment **102**. Similarly, to determine budget information, an image-to-text model fine-tuned for cost-analysis may assess the representations to extract a budget of environment **102**. The extracted budget may in some embodiments be an estimate based on a user profile or the objects or other details of the environment **102**. In some embodiments, a user may provide a budget range. In some embodiments, the budget classifier need not be precise and, in some embodiments, the budget classifier outputs a range of cost.

[0031] At step **310** the processor analyzes the mapping **103** to determine and extract empty or available regions within the space in which an object, such goods and/or services **104**, may fit. For example, the mapping **103** may show empty regions of environment **102**, such as a wall with no decoration or an empty corner. These regions may be available for new goods and services **104**. In some embodiments, the content selection system **100** determines available regions based on determined empty spaces. In some embodiments, for example embodiments in which existing items are replaced, the content selection system **100** determines these regions through characteristics of the regions. In some embodiments, determining available regions includes determining measurement of the environment **102** or the region, as well as other characteristics, such as color, lighting, access to electrical outlets, and any other information that might inform the content item **105** selection process.

[0032] At step **312**, the process **300** then identifies, using a processor (e.g., processor **224**), candidate locations for objects within the free space determined at step **310**. For example, as discussed above, content selection system **100** may select empty regions for placement of goods

and/or services **104**. In some embodiments the content selection system **100** replaces existing items in addition to or instead of determining available, unoccupied, or underutilized regions for new items. These replaceable objects may be selected from a catalog of recognized objects similar to above step **308**. In such embodiments, the system **100** may not perform step **310** as the object replaces an existing object and does not require a new space. When a region's characteristics match those a good or service **104** requires, the content selection system **100** may determine that region to be a candidate for the good or service **104**. For example, if the content selection system **100** recognizes that a living room has only chairs, it might recommend replacing the chairs with a sofa. It might then identify a region of the living room as available for a sofa if the region has the required dimensions for a sofa and does not include a window or a door to avoid blocking these features with the proposed sofa. Other characteristics may also be relevant, such as a window position relative to a screen, internet connection strength for internet connected devices, optimizing walk through flow of a room, style preferences, or budget considerations for fixtures that require plumbing.

[0033] Once the content selection system **100** has determined locations for goods and/or services **104** within the environment **102**, it may then select goods and/or services **104** and create, or retrieve from an outside source, a representation of those goods and/or services **104** in the environment **102**. To select goods and/or services, process **300** first maps goods and/or services **104** from an object storage, such as goods and/or services **104** from object storage **222**, into the locations determined at step **312**.

[0034] At step **316** the process **300** determines if an item matches a location, such as the empty or available space determined at step **310**. An item may match a location if, for example, an associated object or service physically fits the location, meets budget or style requirements of the location, or the location meets the needs of the items, such as providing access to an electrical outlet or limiting UV exposure. If a good and/or service does not match, for example the item does not physically, visually, and/or financially fit, the process **300** returns to step **314** and selects another good and/or service for mapping. If the good and/or service does match, the process moves to step **318** in which the process **300** marks the good and/or service as a potential candidate for advertisement or other representation.

[0035] The process **300** then moves to step **320** to determine if more goods and/or services or locations are available. If the process **300** determines that more goods and/or services or spaces are available, it returns to step **314** to map additional objects. If at step **320** the process determines that more goods and/or services or spaces are not available, it proceeds to send the list of identified goods and/or services to an advertising server, or other supplemental content server, at step **322**. From the advertising or supplemental content server, the content selection system **100** generates a content item **105** for display on primary device **212**, the content item **105** being related to the selected good and/or service.

[0036] In yet another embodiment, the content selection system **100** may also measure the efficiency of the content item **105** showing goods and/or services **104** in relation to a mapping of an environment **102** by monitoring changes to that environment **102** obtained by subsequent scans and mappings **103**. For example, an updated mapping **103** of an environment **102** may show that content item **105** was successful if the related good or service appears in updated mapping **103** but not in the original mapping **103**. Further to that determination, the content selection system **100** may also detect if the location of that new good within the environment **102** matches the location originally forecast for such good. Upon determining that the location is different, the content selection system **100** may adjust its object installation model accordingly. If the location is accurately predicted the object installation model may be promoted within content selection system **100**.

[0037] In some embodiments, the content selection system **100** may also use predictive analysis based on set rules or using a trained model to forecast a future need of an object based on a digital

representation of an environment **102**. For example, after obtaining a first scan of an empty bedroom, then a second scan a few months later of the same bedroom now containing a crib, the content selection system **100** may infer the need for a children's bed and an associated timeline. Such historical information may be stored, in some embodiments, in user profile **228** which include historical information as well as changes to an environment **102**.

[0038] In some embodiments, the content selection system **100** may rearrange existing objects in an environment **102**. In such an embodiment, the content selection system **100** creates a catalog of objects as discussed above and, after obtaining a mapping **103** of the environment **102**, determines alternative locations for the objects. In such embodiments the content item **105** created may be, for example, a new furniture layout floor plan or a three-dimensional, or XR, rendering of the updated arrangement. These embodiments may be useful in rearranging furniture or making space for additional objects.

[0039] In some embodiments, the content selection system **100** may arrange existing objects in a new environment **102**. For example, system **100** may catalog existing objects and place them in a new environment **102**. Such embodiments may be useful when a user moves to a new dwelling and would like to imagine existing furniture or other items in the new home. In such embodiments the content selection system **100** may obtain mappings **103** of the new home from real estate listings or traditional scans.

[0040] FIG. 4 shows an example three-dimensional mapping **103**. In some embodiments, the content selection system **100** may acquire a three-dimensional scan, which may include color imaging information, of a user's room, or environment **102**, when a user uses a VR headset in a room. In some embodiments, the content selection system **100** may acquire a two-dimensional scan when the room is cleaned by a robotic vacuum cleaner. In some embodiments, the content selection system **100** may collect data on multiple related environments **102**. For example, in some embodiments, the content selection system **100** may receive a scan of each room in a house. By collecting scans of an environment **102**, content selection system **100** either receives or generates a mapping **103** of the environment **102** that contains spatial and visual information of the environment **102**.

[0041] Content selection system **100**, in some embodiments, infers adequate locations for goods and/or services **104** based on a mapping **103**. For example, FIG. 4 shows an empty space on the wall on the left. The content selection system **100** may recognize this as an available region **401** and a location for a television or artwork. In some embodiments, content selection system **100** considers additional information from the mapping **103**, scan, or images of environment **102** that might impact location. Such additional information might include lighting, UV exposure, electrical outlet locations, or walkways for example. This information may come from object recognition discussed in step **308** in which content selection system **100** identifies objects, such as sofa **402**, windows **403**, table **404**, and rug **405**. In some embodiments, content selection system **100** infers the goods themselves, deciding goods and/or services **104** that are useful in the environment **102**. For example, the environment **102** shown in FIG. 4 appears to be a living room, which content selection system **100** may recognize through the object recognition of sofa **402**, windows **403**, table **404**, and rug **405**. The content selection system **100** may also identify that the room does not include a television and may suggest such a product to the user. Alternatively, the content selection system **100** may determine that an adjacent room includes a large television and instead suggest artwork for the living room as an additional television is not likely necessary. As discussed above, the content selection system **100** may select the objects, such as goods and/or services **104**, from a storage, such as object storage **222**. The content selection system **100** may select objects that best fit the available locations. In some embodiments, the content selection system **100** infers a design style, such as modern, farmhouse, or maximalist, and selects an object that best matches the identified style. In some embodiments, the content selection system **100** infers a budget range and selects an object that best matches the identified budget. In one embodiment the content selection

system **100** uses mapping **103** and other available data to identify existing objects in the environment **102**, stores those objects in a catalog of one or more recognized objects, and selects products based on those existing objects. For example, content selection system **100** might recognize that a user's living room has a television and based on that information will select artwork, instead of a second television, for placement on an empty space on a wall. In one embodiment the content selection system **100** might recognize that a neighboring room in a house has a television and will select artwork, instead of a second nearby television, for environment **102** in that scenario as well.

[0042] In one example embodiment, content selection system **100** scans a living room and selects a set of coffee tables matching the living room from an object storage **222** containing furniture. As a result, when the user uses an entertainment platform, such as a social media network or Internet-connected TV, that employs content selection system **100** to target consumers with advertisements, the user sees an advertisement for the selected coffee tables instead of a generic advertisement for a generic coffee table. Additionally, if color imaging data is also available and usable, the content selection system **100** may present a content item **105**, such as an advertisement, showing the selected coffee tables installed in the user's own living room. The image may be created by rendering the model in which the new goods and/or services **104** have been inserted into a representation of environment **102**.

[0043] In some embodiments, content selection system **100** determines based on a mapping **103** that a service is adequate for an environment **102**. In an example embodiment, FIG. 5 shows a scan **500** of an environment **102**, a living room with large windows **501**. Based on the large windows **501**, the content selection system **100** may determine that an advertisement of a window cleaning service or a blinds installation company may be adequate. In some embodiments, the content selection system **100** detects via mapping **103** the state of the window **501**, such as the cleanliness of the windows, and generates content items based on that state. For example, content selection system **100** might generate a content item **105** for window cleaning upon recognizing that the windows are dirty, such as content item **503**, which is an image of the environment **102** with **501** professionally cleaned.

[0044] In some embodiments, content selection system **100** includes a user profile **228**, which stores user data, such as preferences, purchase history, history of environment **102**, and other information. Content selection system **100** may use data in user profile **228** to select goods and/or services **104** for environment **102**. For example, data in user profile **228** may indicate a preferred budget and style. Content selection system **100** may then select only goods and/or services that meet that style and budget in addition to physically fitting within the environment **102**.

[0045] The content selection system **100**, in some embodiments, may select a set of products that are relevant to the mapping **103** as there may be more than one product and/or service **104** in the object storage **222** that are relevant to the environment **102**. The content selection system **100** may then rank each product and/or service **104** based on how well each matches the environment **102**. Such determination may include dimensional analysis such as determining that the selected product covers a sizable portion of the detected available installation space within the mapping **103** and not just a small portion. Such determination may also include aesthetic considerations. For example, the content selection system **100** may determine, based on the mapping **103**, that several pieces of furniture detected within the mapping **103** belong to a set and that the set includes another piece of furniture, not present in the mapping **103** but part of the object storage **222**. In some embodiments, the content selection system **100** recognizes a design style of an environment **102**, such as modern, farmhouse, maximalist, or traditional, and selects products, such as furniture or other objects that match that style. For example, content selection system **100** may select a bean-shaped table for an environment **102** with a midcentury modern style, but select a sturdy and traditional wood table for an environment **102** with a colonial style.

[0046] The content selection system **100** may use methods known in the art to generate a three-

dimensional representation of an environment **102**. For example, the content selection system **100** may use photogrammetry methods to reconstruct in three-dimensions the environment **102** based on a series of pictures of the space available, for example, on a social media platform. The content selection system **100** may use semantic segmentation to extract and identify objects from the reconstructed three-dimensional model and match these objects to object storage **222** for which it has full geometric information. The content selection system **100** may also use two-dimensional information to compute space allocation within an environment **102**. For example, a robotic vacuum cleaner session log may be used to infer location of furniture (where the vacuum cleaner cannot go) changes in surface cover (i.e., carpet, rugs, wood floors, tiles, etc.) which may be sufficient to map certain goods (such as new furniture) and services (such as housekeeping services). The content selection system **100** may also detect three-dimensional objects directly from a point cloud capture such as those generated by a VR headset using methods known in the art. Once objects have been detected, the content selection system **100** may then infer free space in the environment **102** using known methods. Once the content selection system **100** determines free space, the content selection system **100** may then further segment that free space based on rooms and areas within rooms. For example, the content selection system **100** may detect an opening created by two shelves and separate it from an opening in front of a sectional sofa, so that different articles can be mapped for these different openings. In this way, the content selection system **100** generates a unique representation of environment **102**.

[0047] In some embodiments, the content selection system **100** may determine that an object present within the mapping **103** may be replaced by an object from the object storage **222**. That determination may be based on recognizing the object in the mapping **103** and determining that the object is past its useful life. In some embodiments, that determination may be based on detecting that the object style does not match the rest of the objects in mapping **103**. In some embodiments, that determination is based on a user profile indicating the age of the object. In some embodiments, the content selection system **100** identifies objects in environment **102** and tracks their age and use. It then stores this information in a user profile. For example, the content selection system **100** may recognize a couch in environment **102**. The content selection system **100** may recognize that two years later, the couch is in a different room. This relocation may indicate to the content selection system **100** that the couch no longer serves its original purpose and should be replaced. The content selection system **100** may then suggest a new couch and display a content item with that suggestion on primary device **212**. In another example, the content selection system **100** may, eight years later, determine that the couch is ten years old, indicating to the content selection system **100** that an updated couch may be more stylish. The content selection system **100** may then suggest a new couch and display a content item with that suggestion on primary device **212**. In another example, content selection system **100** may detect a crib in environment **102** and determine that environment **102** is a baby's bedroom. After two years, the content selection system **100** may detect that the crib remains in environment **102**, and display content items **105** related to toddler beds, knowing that children often transition out of cribs sometime after two years and that the crib is likely to soon be replaced. Upon determining that the object may be replaced by an object from the object storage **222**, the content selection system **100** may present a content item **105** featuring the object such as an advertisement for that replacement object. The content selection system **100** may also include a stock photograph or video of the old product as part of the content item to further personalize the content item. For example, an advertisement message might read, "Your old couch needs a refresh! Here's a new one!" while showing the old couch in the first portion of the advertisement and the new couch in a second portion. Methods such as the Visual Complements Model may be used to determine the visual appeal of an object within an environment **102** or as a replacement for an existing object within a set of objects and hence compute a desirability score for a particular object within a set and generate a content item **105** accordingly. In some embodiments, the generation of 2D images of the environment **102** using virtual cameras at different angles of the room may also

contribute to determining the visual appeal of an object.

[0048] FIG. 6 shows an illustration of an example embodiment process of content selection system **100**. The content selection system **100** obtains a mapping **601** of a user's living room, an environment **102**, from a secondary device **210** such as an XR headset or a robotic vacuum cleaner. In the example shown in FIG. 6, the content selection system **100** generates for display a 3D rendering of the mapping **601** on, for example, a television screen where the television is a primary device **212**. In similar embodiments, the content selection system **100** may display rendering on other devices such as a computer screen, smartphone, or XR headset. Next, the content selection system **100**, using mapping **601**, analyzes goods and/or services **104** in an object storage **222** and identifies objects relevant to the environment **102**. In the embodiment shown in FIG. 6, the objects in objects storage **222** include images of household items. Relevant goods and/or services **104** may be goods and/or services **104** that meet an identified need, physically fit the environment **102**, or match a style of the environment **102**. At this step content selection system **100** also determines that some goods and services **104** are not relevant. For example, the content selection system **100** identifies some goods and/or services **104** such as those in the coffee table and bookshelf categories as not relevant because the mapping **601** includes no location in which these objects can physically fit. The content selection system **100** determines, either as part of this analysis or after, an adequate location **602** for a series of objects. In the illustrated embodiment, the location **602** is an empty space on a wall of the living room that the content selection system **100** has identified using mapping **601**. After identifying location **602**, the content selection system **100** determines, for example, that products in the vintage TV category are too thick to fit in the identified installation zone because the installation zone is on a wall and requires a wall mounted television. The content selection system **100** further determines that items in the vertical art category also do not fit as they are small and only fill a limited portion of the identified installation zone, location **602**. The content selection system **100** then determines that items in the horizontal art and LCD TV categories are prime candidates for the scanned space based on their utility and size. The content selection system **100** may then further refine these candidates based on data such as dimensions or price. The content selection system **100** may also rank the candidates using other information, such as user preferences, to further reduce or organize the list of potential candidates of goods in object storage **222** to only TV's most likely to fit the environment **102**.

[0049] In the illustrated embodiment of FIG. 6, the content selection system **100** determines that the LCD TV is the best candidate. After having generated a list of best or selected TV's, extracted from object storage **222**, that match the mapping **601**, the content selection system **100** may then push a content item **603**, here an image depicting the TV in the 3D rendering of the room, to a primary device **212**. In similar embodiments, the content item **603** may be an AR overlay displayed on top of an image of a room on an AR display. In another embodiment, it may be a 3D depiction of an object in a VR environment. In some embodiments, the content item **603** is an advertisement and the image the TV in the room is integrated into the advertisement to create a personalized experience.

[0050] In some embodiments, the content selection system **100** may detect that a secondary device **210** is in a visited environment **102**, such as a resort or an Airbnb, by recognizing that a scan is not of the typical surroundings, a stored environment, an existing 3D rendering, or any other mapping environment. Upon detection that the device is not at a usual living place, the content selection system **100** may also detect goods and/or services **104** at the visited place that the user may be interested in. For example, the user may have taken pictures of a decorative lamp, a dresser or a couch at the visited environment **102**. The content selection system **100** may then recognize at least one object in these pictures as part of an object storage **222** and may also determine that such an object may fit within the usual living quarters using the scans or mappings **103** of the usual living quarters as described in the previous embodiments of this disclosure. In some embodiments, the introduction of a new object may prompt the content selection system **100** to determine and display

an updated layout of environment **102**. In some embodiments, the system **100** determines that the object may replace an existing object in environment **102**. In some embodiments, the system **100** may use similar techniques to rearrange existing objects in environment **102**, for example to update the layout of a room. In these embodiments, the content selection system **100** may rearrange objects based on the theme, style, or look and feel of an environment **102**. Upon determining that at least one object in the object storage **222** fits the living quarters, the content selection system **100** may then elect to generate a content item **105** related to that object on a primary device **212**. In some embodiments, the system **100** displays the content item **105** upon return of the device **210** to the usual living place to improve efficiently and the likelihood that the user positively receives the content item.

[0051] The content selection system **100** may also detect that another secondary device **210** of a different, second user is visiting the same visited environment **102** as the user and device **210** discussed above, but at a later date. In such embodiments the content selection system **100** may also upon determining that that object fits within the living quarters of the second user and second device **210**, present a content item for the same object to the second user. The selection of the content item may, in some embodiments, be based on a mapping **103** of said living quarters, without the second device having to explicitly indicate interest in that object.

[0052] For example, FIG. 7 shows a example process **700** of content selection system **100** illustrating variations on process **300**. The process **700** may be implemented, in whole or in part, by control circuitry including for example, processor **224**. For example, one or more of the aforementioned devices, such as **210**, may execute one or more instructions or routines stored to memory or storage of a device to implement, in whole or in part, the process **700**. In some embodiments, portions of process **700** are implemented on a cloud server. In some embodiments, all steps **700** occur on one device.

[0053] At step **701** of process **700**, content selection system **100** extracts environment information from a received mapping data of an area. The environment information may inform content selection system **100** regarding the environment **102**, such as the layout, style, design, dimensions, objects, or other information. It may include spatial information and/or other data. Environment information may include, for example, at least one of available space, furniture arrangement, a catalog of recognized objects in the area, and/or placements of recognized objects in the area. Mapping data may be information regarding a mapping **103** of environment **102**. For example, mapping data may be the mapping **103** itself or data from which content selection system **100** may create a mapping **103**. In some embodiments, the content selection system **100** creates the mapping based on received data, such as scans or images of environment **102**. Such scans and images may be obtained from device **210** using, for example, a camera or other sensor capable of capturing data related to environment **102**. In some embodiments, the content selection system **100** does not create mapping **103** but instead receives the mapping **103** from another device. The content selection system **100** may receive the mapping data to input/output circuitry **226** from a secondary device **210**, such as a camera, smartphone, or other device capable capturing data, a database, or other source. In some embodiments, content selection system **100** determines environment information from the mapping data using processor **224**. In some embodiments, mapping data includes embedded data containing the environment information. In some embodiments, the environment information is determined based on a combination of receiving data and analyzing the data.

[0054] At step **702** the content selection system **100** accesses via a network **201a** storage of supplemental content items such as object storage **222**. The supplemental content source may be a third-party content source, such as an image or advertising database or service. In some embodiments, the supplemental content source includes a collection of content items **105** for selection. In some embodiments, each supplemental content item of the supplemental content source highlights a specific good or service, such as furniture, artwork, electronics, window cleaning, home remodeling, etc. The content items **105** may be images or advertisements

containing the highlighted good or service.

[0055] At step **703** the content selection system **100** accesses from a profile data source, a user profile **228**. The profile data source may be, for example, a database of user profiles, a memory of a user device, or a user account stored on a cloud server. The user profile may include information such as user preferences, history of objects and other details of environment **102**, or other information. System **100** may continually update the user profile **228** upon receipt of new data, such as user settings updated mappings **103**.

[0056] At step **704** content selection system **100** uses processor **224** to determine, based on the environment information, the storage of supplemental content items, and the user profile **228**, a selected supplemental content item **105** from the storage of supplemental content items matches the environment **102**. This selection may represent for example content item **105** that best addresses a deficiency in an environment **102** or that best fits the dimensions of environment **102** or an available space in environment **102**. The selection may further reflect preferences such as style and budget.

[0057] At step **705** the content selection system **100** in response to the determining the selected supplemental content item matches environment **102**, generates, using processor **224** and input/output circuitry **226**, for display on a display device **212**, the supplemental content item **105**. For example, the content selection system **100** may select a content item **105** advertising a certain TV, such as the embodiment seen in FIG. **6**. The content selection system **100** then displays that advertisement on a user device **212** such as a phone, television, or computer for a user to receive.

[0058] At step **706** the content selection system **100** in response to the determining the selected supplemental content item does not match environment **102**, ends the process. In some embodiments, after **706** the process **700** may return to step **704** to select a second supplemental content item for consideration.

[0059] The processes described above are intended to be illustrative and not limiting. One skilled in the art would appreciate that the steps of the processes discussed herein may be omitted, modified, combined, and/or rearranged, and any additional steps may be performed without departing from the scope of the disclosure. More generally, the above disclosure is meant to be exemplary and not limiting. Only the claims that follow are meant to set bounds as to what the present disclosure includes. Furthermore, it should be noted that the features and limitations described in any an embodiment may be applied to any other embodiment herein, and flowcharts or examples relating to an embodiment may be combined with any other embodiment in a suitable manner, done in different orders, or done in parallel. In addition, the systems and methods described herein may be performed in real time. It should also be noted that the systems and/or methods described above may be applied to, or used in accordance with, other systems and/or methods.

Claims

1. A method comprising: extracting environment information from received mapping data of an environment, wherein the environment information includes at least one of available space, a catalog of one or more recognized objects in the environment, or placement of the one or more recognized objects in the environment; accessing a storage of supplemental content items; identifying, from a profile data source, a user profile; selecting, based on the environment information and the user profile, a supplemental content item from the storage of supplemental content items; and generating, for display on a display device, the supplemental content item.
2. The method of claim 1, wherein selecting the supplemental content item comprises determining that an object related to the selected supplemental content item physically fits within the environment.
3. The method of claim 1, further comprising receiving via a first device the received mapping data, wherein the first device and the display device are different devices.

4. The method of claim 1, wherein the selected supplemental content item is an image or video of an object or service related to the supplemental content item.
5. The method of claim 1, further comprising receiving style information of the environment, and wherein selecting the supplemental content item is further based on the style information of the environment.
6. The method of claim 1, further comprising receiving historical information of the environment, and wherein selecting the supplemental content item is further based on the historical information of the environment.
7. The method of claim 1, wherein the received mapping data of the environment includes a three-dimensional scan of the environment.
8. The method of claim 1, wherein the user profile further includes indicated interest in an object, and wherein selecting the supplemental content item is further based on the indicated interest in an object.
9. The method of claim 1, further comprising extracting environment information from a received mapping data of a second environment, wherein selecting the supplemental content item is further based on the environment information of the second environment.
10. The method of claim 1, wherein the selecting, based on the environment information and the user profile, a supplemental content item is further based on a field of view of the display device, and wherein the environment extends beyond the field of view.
11. The method of claim 10, wherein the supplemental content item is a representation of an object; wherein the generating, for display on a display device, the supplemental content item further comprises generating for display the representation of the object in the field of view; and wherein the selecting of the supplemental content item is further based on recognized objects in the environment outside of the field of view.
12. A system comprising: control circuitry configured to: extract environment information from received mapping data of an environment, wherein the environment information includes at least one of available space, a catalog of one or more recognized objects in the environment, or placement of the one or more recognized objects in the environment; access a storage of supplemental content items; identify, from a profile data source, a user profile; select, based on the environment information and the user profile, a supplemental content item from the storage of supplemental content items; and generate, for display on a display device, the supplemental content item.
13. The system of claim 12, wherein to select the supplemental content item is by determining that an object related to the selected supplemental content item physically fits within the environment.
14. The system of claim 12, the control circuitry further configured to receive via a first device the received mapping data, wherein the first device and the display device are different devices.
15. The system of claim 12, wherein the selected supplemental content item is an image or video of an object or service related to the supplemental content item.
16. The system of claim 12, the control circuitry further configured to receive style information of the environment, and wherein to select the supplemental content item is further based on the style information of the environment.
17. The system of claim 12, wherein the control circuitry is further configured to receive historical information of the environment, and wherein to select the supplemental content item is based on the historical information of the environment.
18. The system of claim 12, wherein the received mapping data of the environment includes a three-dimensional scan of the environment.
19. The system of claim 12, wherein the user profile further includes indicated interest in an object, and wherein selecting the supplemental content item is further based on the indicated interest in an object.
20. The system of claim 12, the control circuitry further configured to extract environment

information from a received mapping data of a second environment, wherein to select the supplemental content item is based on the environment information of the second environment.
21-55. (canceled)
